1.1 Python Data Structures - Lists and Tuples

- How to store three product names inside a single variable?
- Please put the code inside the new cell
- what is a list and how does it store ordered items?
- can you show the third item in the list?
- how did you get the third product? Explain the syntax
- we have a new product “Pants”. add it to the list
- add pants more programmatically without altering the original list
- other ways to add elements in a list
- retrieve the last two products in the list
- can you explain the syntax of the code you wrote?
- does your code work for special cases where the list might be too long or even empty?
- are there any ways to store product categories like Electronics and Clothing such that they would not be changed
- in a new cell create my product categories
- what happens if we try to modify an element in out product categories?

1.2 Python Data Structures - Dictionaries

- can you put the product names and prices inside a container so that i can pass a product name and get its price
- can you explain how you took the name of the product and gave the price?
- in a new cell can you add a new product. iPod priced at 200
- can you explain how you add new items to a dict?
- can you write the code so that the user types in a product name and gets the price?
- what happens if a user puts a product name that does not exist in the dict? can we access a key that does not exist?

1.3 Looping in Python

- we have a list of products. Can you write the code to go through the list and print out the name of each product?
- can you explain how the loop works?
- can you modify the code so that product name and a counter is displayed?
- Write the code that initially has an empty list. Then it keeps asking the user for new product names, and adds them to the list. it will stop when the user inputs exit
- explain how while handles adding new products to the list
- how does your code stop the infinite loop?

- write new code that has an empty list initially. then asks user for prices. checks each price to be positive. If negative it does not add it to the list. if zero, it stops asking for prices. and if positive, it adds it to the list.
- how does you code handle negative prices?
- how does your code end?
- can you explain the difference between continue and break?
- check the code infinite loops
- fix this potential bug

1.4 Functions in Python

- i am calculating the discount for two prices. if the total is greater than 1000, 10% discount is applied. The final price is then shown to the user. Can you make my code reusable so I don't have to write it every single time?
- can you tell me what def and return are
- You are an experienced Python developer.

  I am learning python and currently working with a billing app. IT calculates the total price based on two input prices. then it applies a discount.

  If the total is above 1000 then discount is 10 percent. if not the discount is 0.

  give me a reusable code for my app

- add calculate_bill to a new cell
- what happens when the user inputs a single price?
- in the new calculate_bill function, what happens to price2?